What role could decision models play in GeoAI?

Qualifying Spatial Relations using Jev.

Disposition for a master thesis by Jakob Roland
Supervised by Prof. Dr. Krzysztof Janowicz
University of Vienna, 2026

Introduction

Janowicz et al., 2020 asked if there will be an *"artificial GIS analyst"* by 2030. Since then there have been huge strides both in the field of artificial intelligence (AI) and autonomous geoinformation systems (GIS) (Li et al., 2025). As spatial analysis requires large amounts of up-to-date geodata, GeoAI (Janowicz et al., 2020) requires substantial compute, leading to low accessibility and high environmental impact (Janowicz, 2023; Shi et al., 2025). Also, large language models (LLMs) may produce a high overhead in reasoning tokens with high latency (Chen et al., 2025; F. Duan et al., 2026; Wang et al., 2026) and a long-form response in natural language, which might be difficult to process downstream. *Decision models*, also described as *system-one* models (Kahneman, 2011), might be more suitable for certain GeoAI applications, as they operate more efficiently and quickly than LLMs, as well as responding in a predetermined, consistent format. To test this assumption, this thesis reproduces the experiment on *spatial relation qualification* done by Ji et al., 2025 using Typesafe AI's model Jev (Almeida, 2026) to enable a direct comparison of the two models.

Research Goals

The aim of this thesis is to do an initial exploration of the model Jev, in the hopes of gaining broader insights into decision models in spatial settings. As Jev is not a chat-based model, proven prompt engineering methods, like few-shot learning or chain-of-thought reasoning (Schulhoff et al., 2025), might not have the same advantages as with LLMs. This will be tested using the same prompting strategies as in the original experiment. Ethical points, especially sustainability, but also briefly bias (Aldahoul and Zaki, 2026), will be discussed as well. In total, the thesis links up with works about GeoAI (Janowicz et al., 2020; Ji et al., 2025), autonomous GIS (Li et al., 2025), and ethical use of GeoAI (Janowicz, 2023; Shi et al., 2025). The thesis also wants to contribute to works evaluating and benchmarking Jev (Deußer et al., 2026; F. Duan et al., 2026; Wang et al., 2026).

The following research questions will be addressed:

- How does the decision model Jev compare with ChatGPT-4 in the task of spatial relation qualification in terms of accuracy and cost?
- In what ways could decision models benefit GeoAI, especially considering sustainability and process integration?
- Does few-shot prompting and chain-of-thought reasoning improve results for system-one intelligence models?

Methodology

To ensure a good comparison, the thesis will stay as true as possible to the original experiment in Ji et al., 2025, using the same data and setup. The prompts will have to be adapted to the unique requirements of Jev, but the same strategies (few-shot prompting and chain-of-thought reasoning) will be tested. The experiment will utilize Typesafe AI's API and Python SDK to access Jev 1.13.0. All code and results will be made available together with the thesis on GitHub. A short review of related works will accompany the experiment and discussion.

Expected Results and Challenges

Initial experimentation proved Jev's speed and (cost-)efficiency, while it could not beat the benchmark set by ChatGPT-4 in terms of accuracy. Still, the confidence score, which Jev assigns to its answers, seems reliable, and some interesting patterns occurred in the initial results, which merit further investigation.

The main challenge this thesis will face is a problem common with all LLMs: Their opacity. Jev's answers lack any reasoning and cannot be reproduced reliably. In addition, Jev is a proprietary model, with very little official information released publicly.

References

Aldahoul, N., & Zaki, Y. (2026, October 19). *The Jev Decision Model Applies Astrological And Demographic Stereotypes When Recommending Friends*. https://doi.org/10.2139/ssrn.7633902

Almeida, D. (2026, September 15). *Introducing System One Models & Jev*. Retrieved October 23, 2026, from https://typesafe.ai/blog/introducing-system-one-models-and-jev

Chen, X., Xu, J., Liang, T., He, Z., Pang, J., Yu, D., Song, L., Liu, Q., Zhou, M., Zhang, Z., Wang, R., Tu, Z., Mi, H., & Yu, D. (2025). Do NOT Think That Much for 2+3=? On the Overthinking of Long Reasoning Models. *Proceedings of the 42nd International Conference on Machine Learning*, 9487–9499. Retrieved October 24, 2026, from https://proceedings.mlr.press/v267/chen25bx.html

Deußer, T., Sparrenberg, L., & Sifa, R. (2026, September 29). *Evaluating and Benchmarking the System One Model Jev*. arXiv: 2609.37647 [cs.CL]. https://doi.org/10.48550/arXiv.2609.37647

Duan, F., Lin, J., Wang, J., Xiang, J., Wu, J., Zhang, X., Yan, H., Wang, S., & Wei, Z. (2026, October 12). *General Decision Models: Benchmarking and Insights Beyond Jev*. arXiv: 2610.03935 [cs.CL]. https://doi.org/10.48550/arXiv.2610.03935

Duan, W. (2020). Automatic alignment of contemporary vector data and georeferenced historical maps using reinforcement learning. *International journal of geographical information science : IJGIS*, *34*(4), 824–849.

Janowicz, K. (2023, March 27). *Philosophical Foundations of GeoAI: Exploring Sustainability, Diversity, and Bias in GeoAI and Spatial Data Science*. arXiv: 2304.06508 [cs.CY]. https://doi.org/10.48550/arXiv.2304.06508

Janowicz, K., Gao, S., McKenzie, G., Hu, Y., & Bhaduri, B. (2020). GeoAI: Spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. *International Journal of Geographical Information Science*, *34*(4), 625–636. https://doi.org/10.1080/13658816.2019.1684500

Jev 1.13 jaggedness. (n.d.). TypeSafe. Retrieved October 26, 2026, from https://docs.typesafe.ai/model-jaggedness/jev-1.13

Ji, Y., Gao, S., Nie, Y., Majić, I., & Janowicz, K. (2025). Foundation models for geospatial reasoning: Assessing the capabilities of large language models in understanding geometries and topological spatial relations. *International Journal of Geographical Information Science*, *39*(9), 1866–1903. https://doi.org/10.1080/13658816.2025.2511227

Kahneman, D. (2011). *Thinking, fast and slow* (1st ed). Farrar, Straus and Giroux.

Li, Z., Ning, H., Gao, S., Janowicz, K., Li, W., Arundel, S. T., Yang, C., Bhaduri, B., Wang, S., Zhu, A.-X., Gahegan, M., Shekhar, S., Ye, X., McKenzie, G., Cervone, G., & Hodgson, M. E. (2025). GIScience in the era of Artificial Intelligence: A research agenda towards Autonomous GIS. *Annals of GIS*, *31*(4), 501–536. https://doi.org/10.1080/19475683.2025.2552161

Schulhoff, S., Ilie, M., Balepur, N., Kahadze, K., Liu, A., Si, C., Li, Y., Gupta, A., Han, H., Schulhoff, S., Dulepet, P. S., Vidyadhara, S., Ki, D., Agrawal, S., Pham, C., Kroiz, G., Li, F., Tao, H., Srivastava, A., … Resnik, P. (2025, February 26). *The Prompt Report: A Systematic Survey of Prompt Engineering Techniques*. arXiv: 2406.06608 [cs]. https://doi.org/10.48550/arXiv.2406.06608

Shi, M., Janowicz, K., Verstegen, J., Currier, K., Wiedemann, N., Mai, G., Majic, I., Liu, Z., & Zhu, R. (2025). Geography for AI sustainability and sustainability for GeoAI. *Cartography and Geographic Information Science*, *52*(4), 331–349. https://doi.org/10.1080/15230406.2025.2479796

Wang, Z., Xue, Z., Mu, J., Liu, H., Yuan, Y., Wu, H., & Li, L. (2026, October 19). *Truncated Reasoning Under a Budget: Evaluating System-1 Models as Decision-makers*. arXiv: 2610.25768 [cs.CL]. https://doi.org/10.48550/arXiv.2610.25768

scratchpad

Struggle with mathematical accuracy Deußer et al., 2026; F. Duan et al., 2026; “Jev 1.13 Jaggedness”, n.d.

Long context degrade results W. Duan, 2020; “Jev 1.13 Jaggedness”, n.d.